

✓ RNN → STM
✓ LSTM → HS
CS ← Complex

⇓
GRU

ChatGPT

BERT

Google

Transformer Architecture

⇓
RNN, LSTM

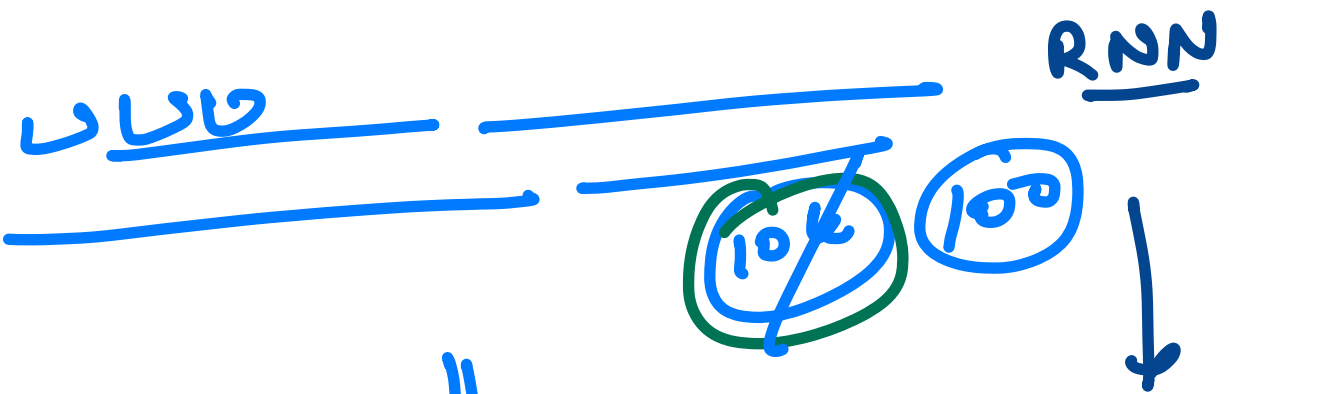
Theory

Wix

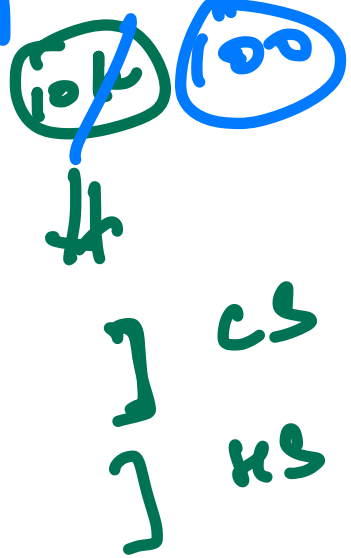
LLM

App

LLM

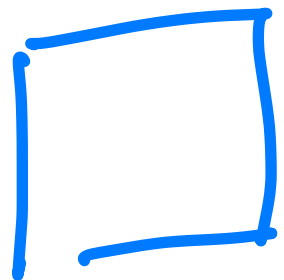


long term
memory

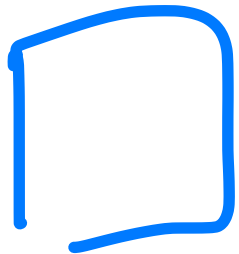


LSTM

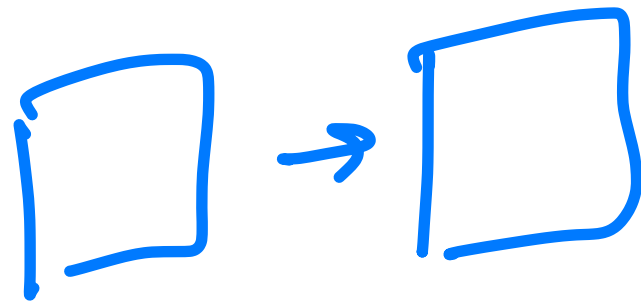
?



Time Step 1



Time Step 2



Running

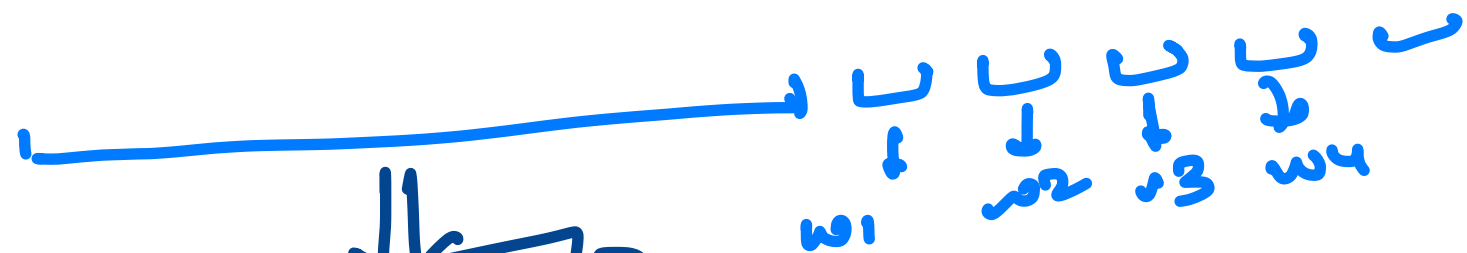
Running

wow!

wow + !

unbelievable

un + believe + able

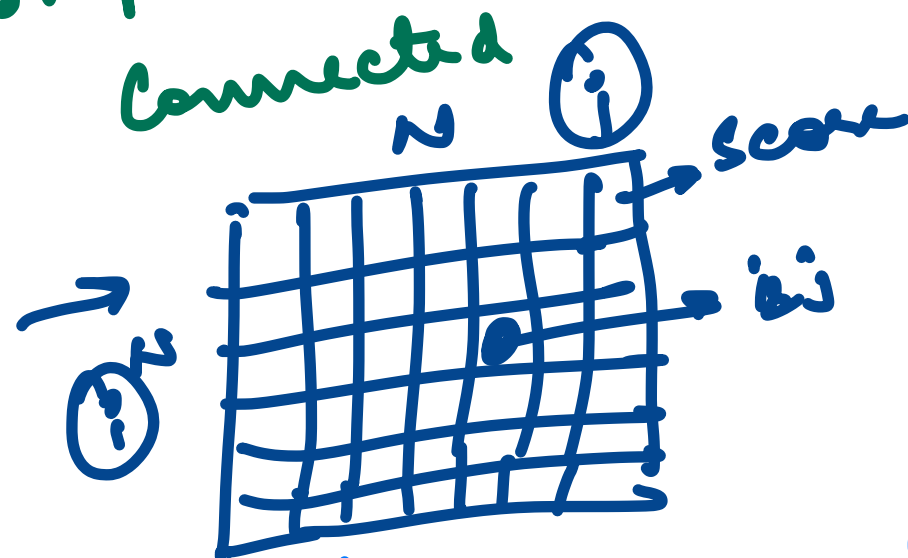


? Parallel?

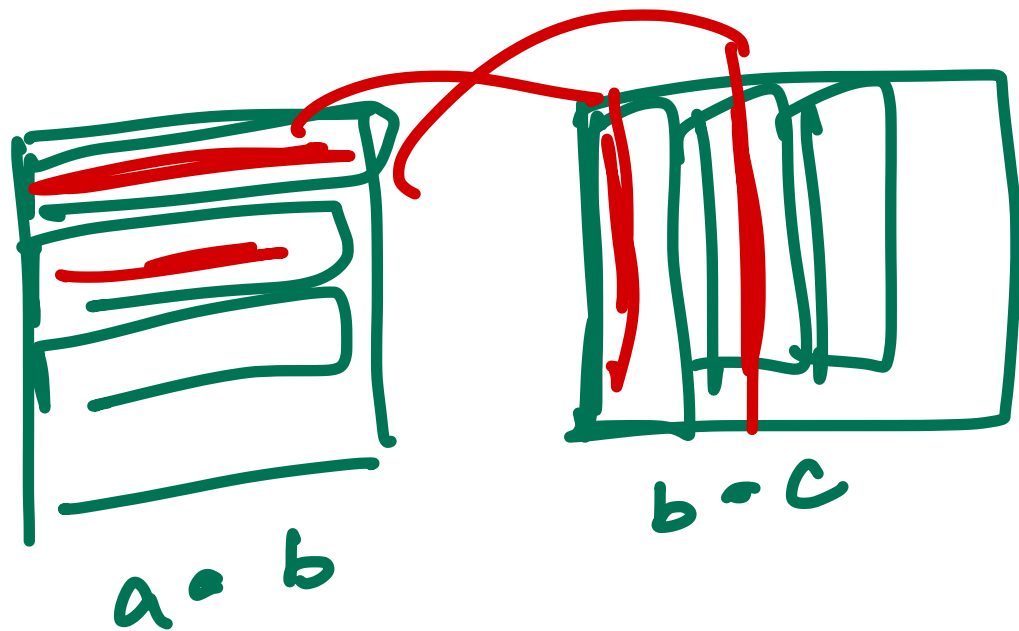
N Tokens $\rightarrow$ Relationship /
Meaning /
Context /
Impact /

N Tokens Attending to

N^2 Tokens



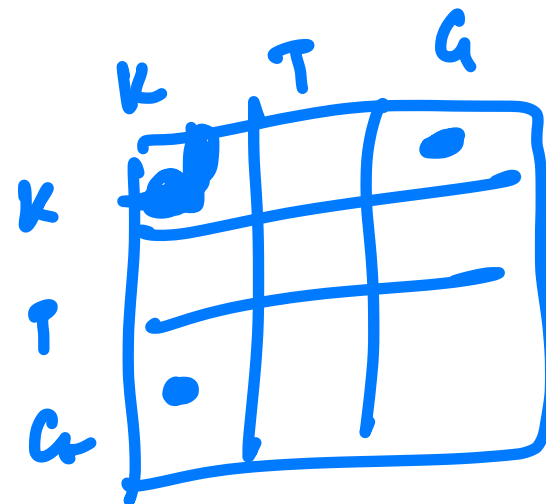
Self Attention Matrix



Parallel
computation

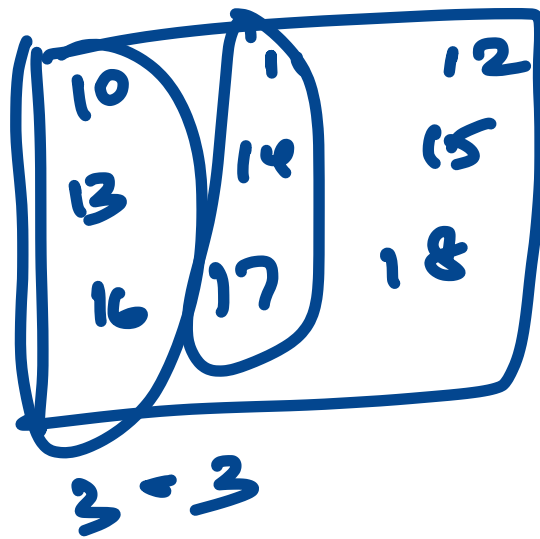
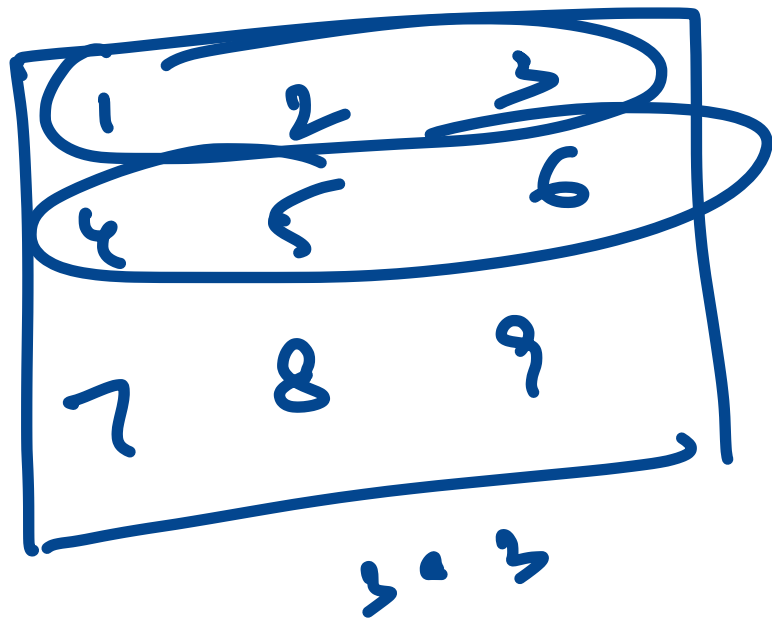
Problem with this parallel computation?

Sequence, order

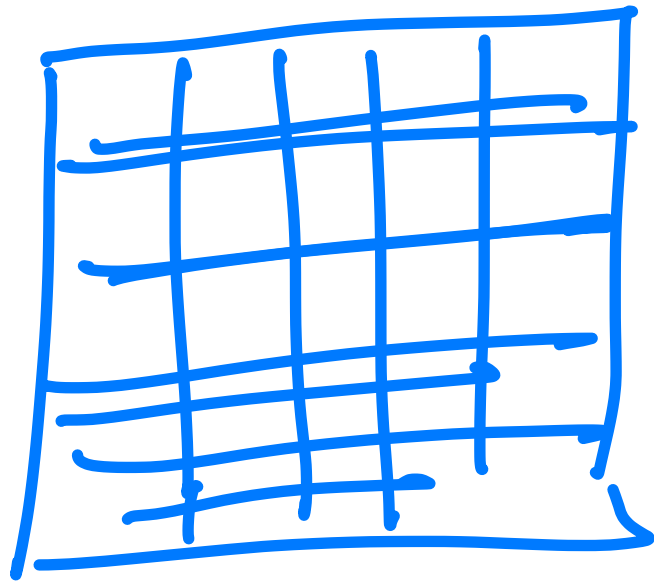


Keerti teaches Govtanny
Govtanny teaches Keerti

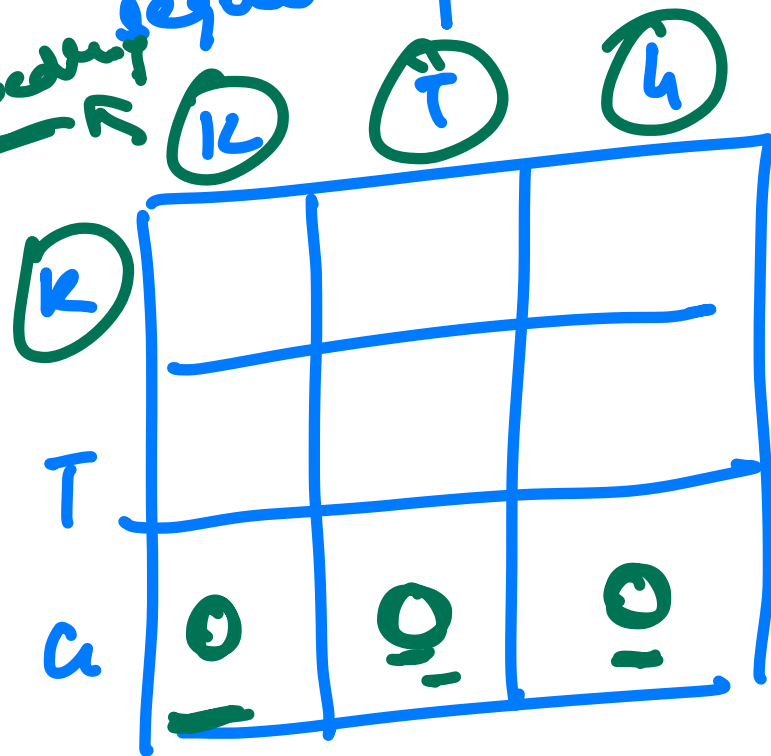
{ Cat sat on the mat } → (A Nabin)
 Mat sat on the cat



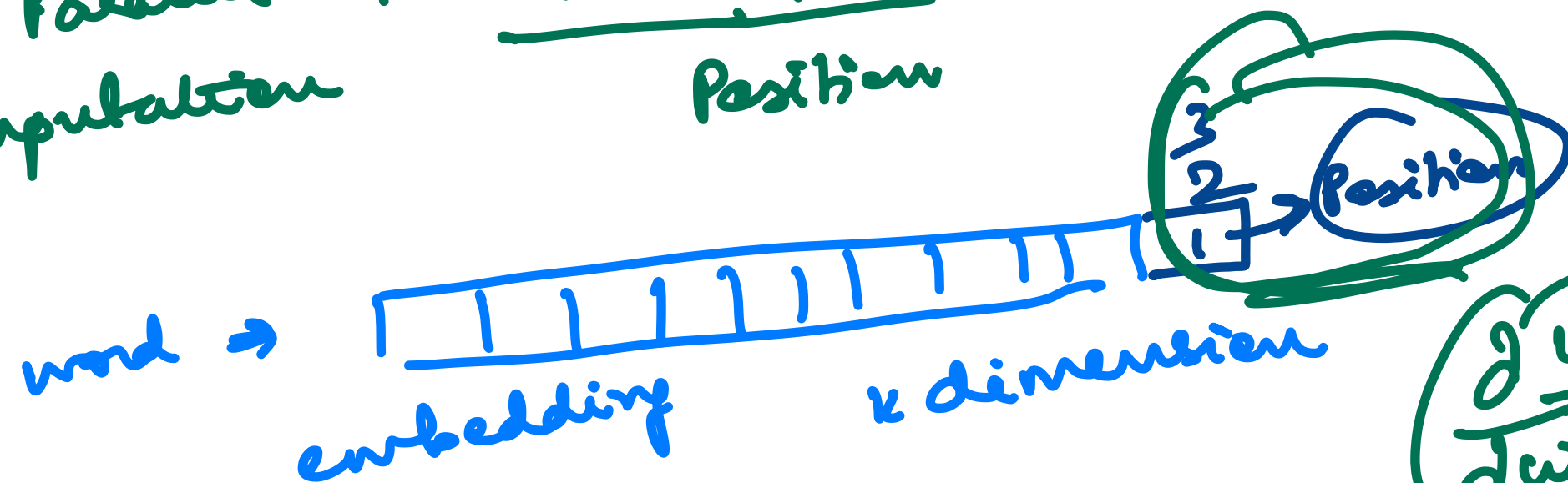
$$\begin{aligned}
 &1 = 10 + 2 = 13 + 3 = 16 \\
 &9 = 1 + 5 = 14 + 6 = 17
 \end{aligned}$$



last the
embedding sequence order



Parallel + Sequence / order
Computation Position



$$\frac{\partial \text{loss}}{\partial w}$$

$\downarrow$

cost

cat sat on the mat

Keete
①

Teacher
②

Location
③

Neural Network $\rightarrow$ Gradients

Continuous functions

Dimension $\rightarrow$ weight

$\rightarrow$ 2, 3
discrete

$\rightarrow$ computation

Keerti teaches at Educosys

vocab $\rightarrow 4$ words
Embedding $\rightarrow 4$

			0.6	2.3	
{	1.2	1.1			→ embedding of teaches
		0.7	0.3	0.2	
{	0.8				→ embedding of at
{					
{					→ Embedding
{					

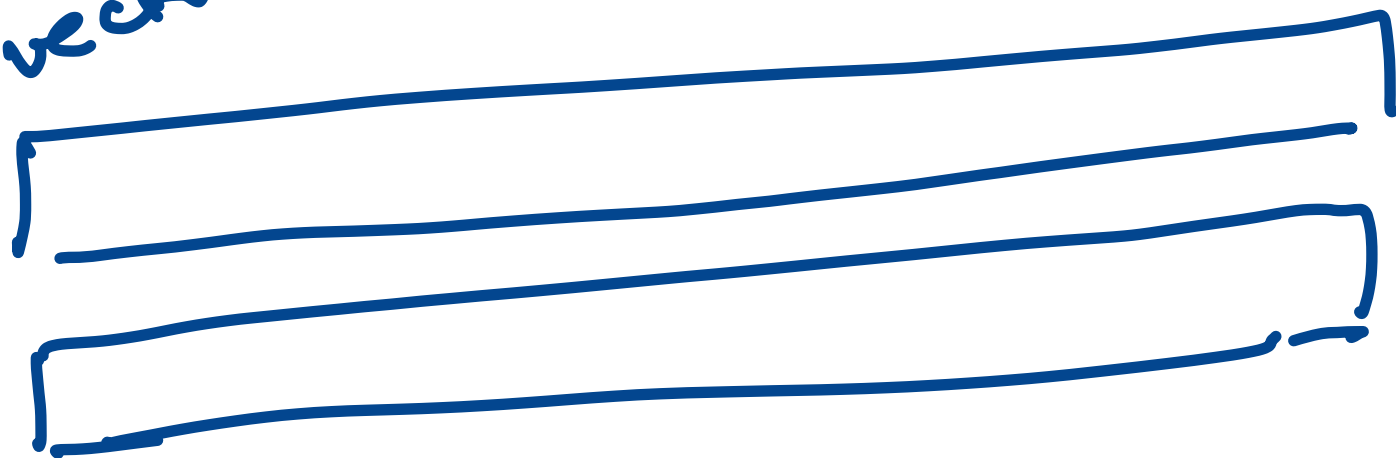
Tokenization $\rightarrow$ encoding $\rightarrow$ embedding
 $\downarrow$
word $\rightarrow$ vocab
 $\downarrow$
content
Static embedding.

k dimensions

Bank $\Rightarrow$ River Bank
Many Bank
Apple $\Rightarrow$ company
Fruit

Position Encoding $\Rightarrow$ Intuition
Makes formula

vector



position 0

1

2

3

4

...

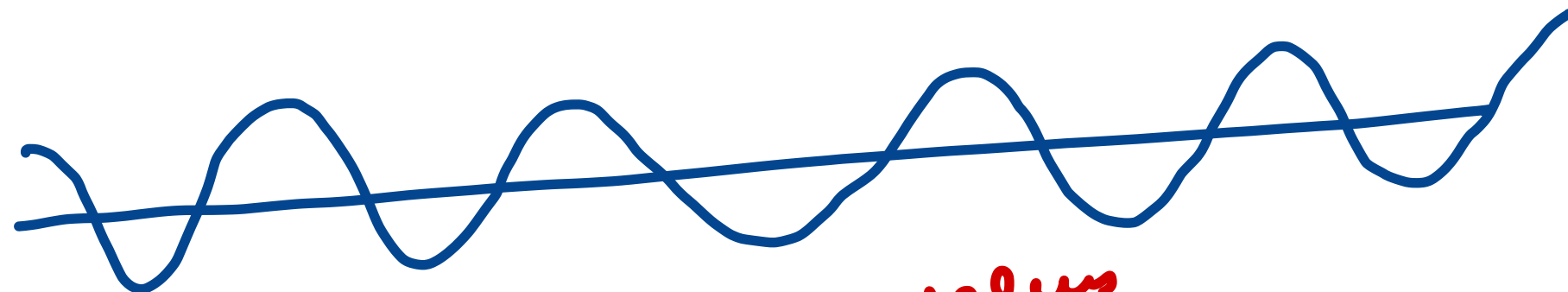
1

100

100 position

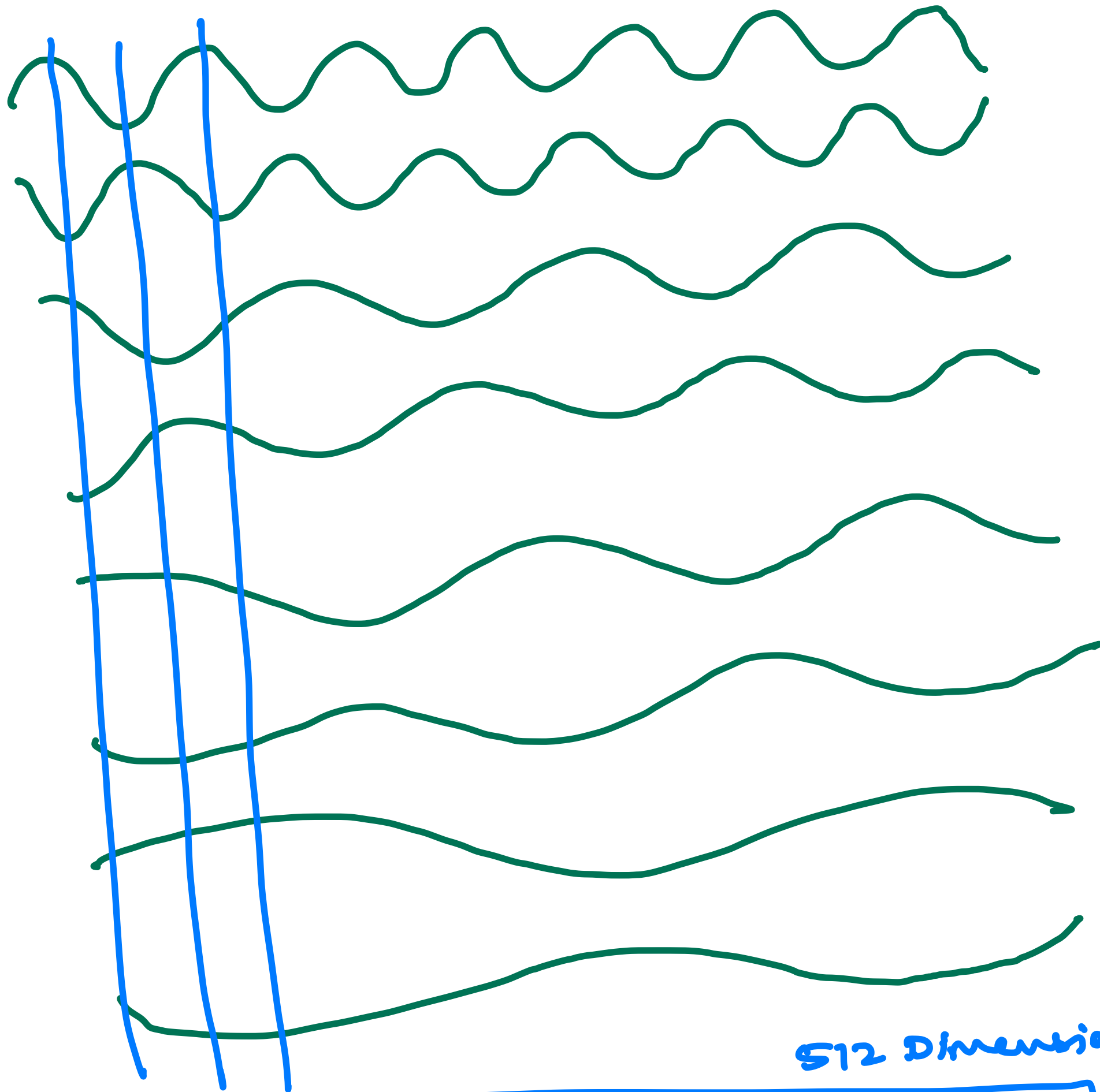
→ Each position should have unique encoding

→ Continuous



$\sin + \cos \Rightarrow$ some values
unique

after a
while $\Rightarrow$ repeat



512 Dimensions
encoded

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

$$pos=1 \quad \sin 1 \quad \cos 1 \quad \sin\left(\frac{1}{10000^{2i/d}}\right) \quad \cos\left(\frac{1}{10000^{2i/d}}\right) \quad \sin\left(\frac{1}{6.3}\right) \cos\left(\frac{1}{6.3}\right) \dots$$

$$\sin 1 \quad \cos 1 \quad \sin\left(\frac{1}{2.5}\right) \cos\left(\frac{1}{2.5}\right) \quad \sin\left(\frac{1}{6.3}\right) \cos\left(\frac{1}{6.3}\right) \dots$$



input embedding

+

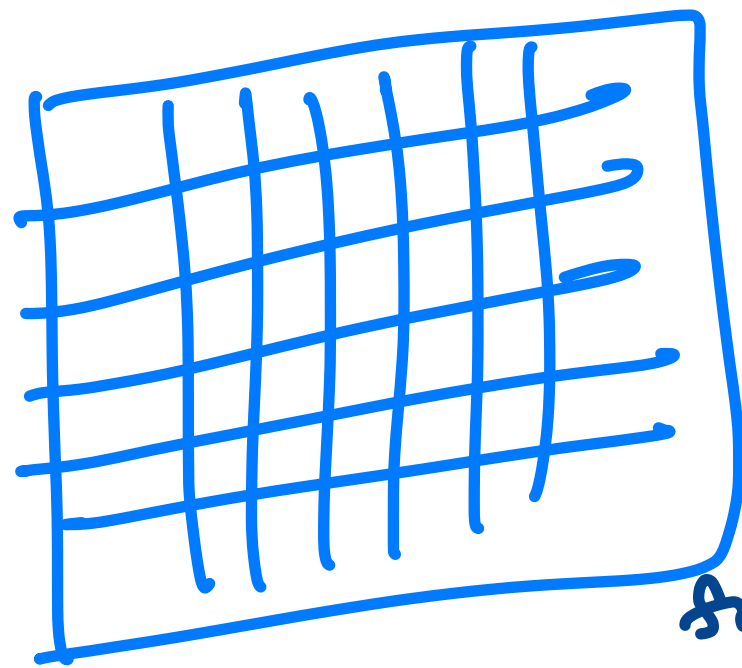


positional encoding

"



= k dimension

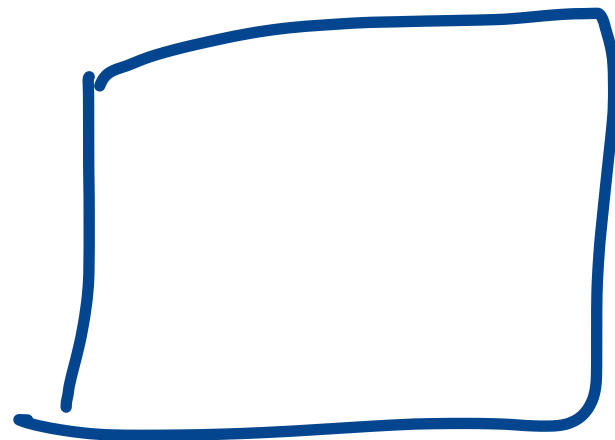


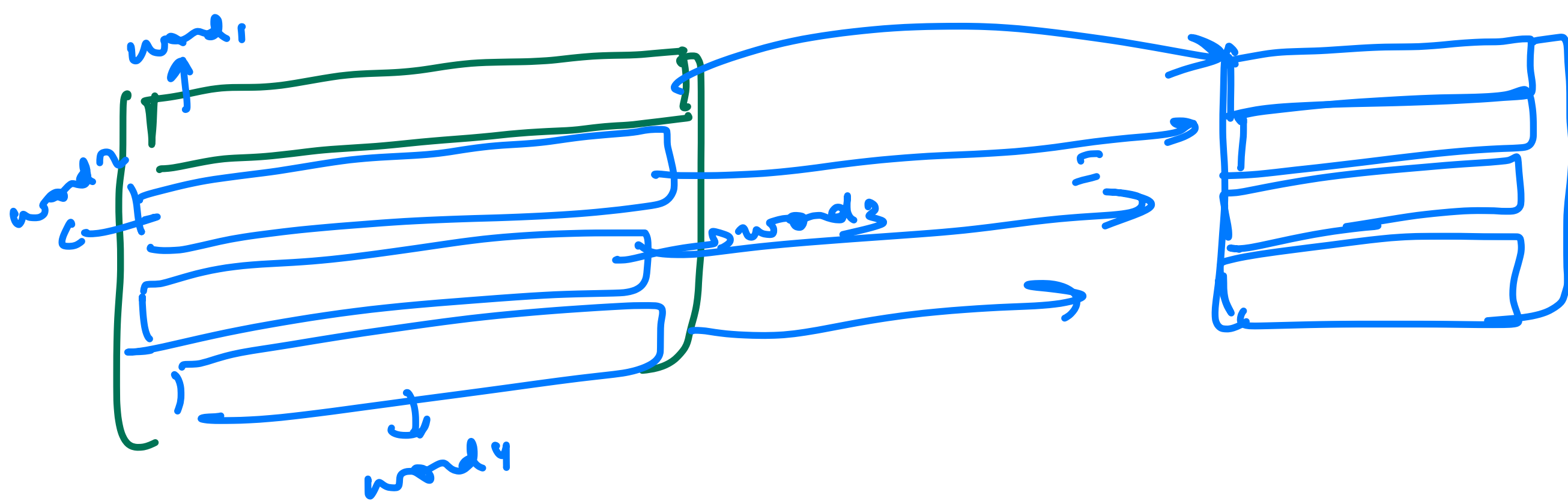
Understand
Relationship between
words?

I love eating ~ apples in the morning

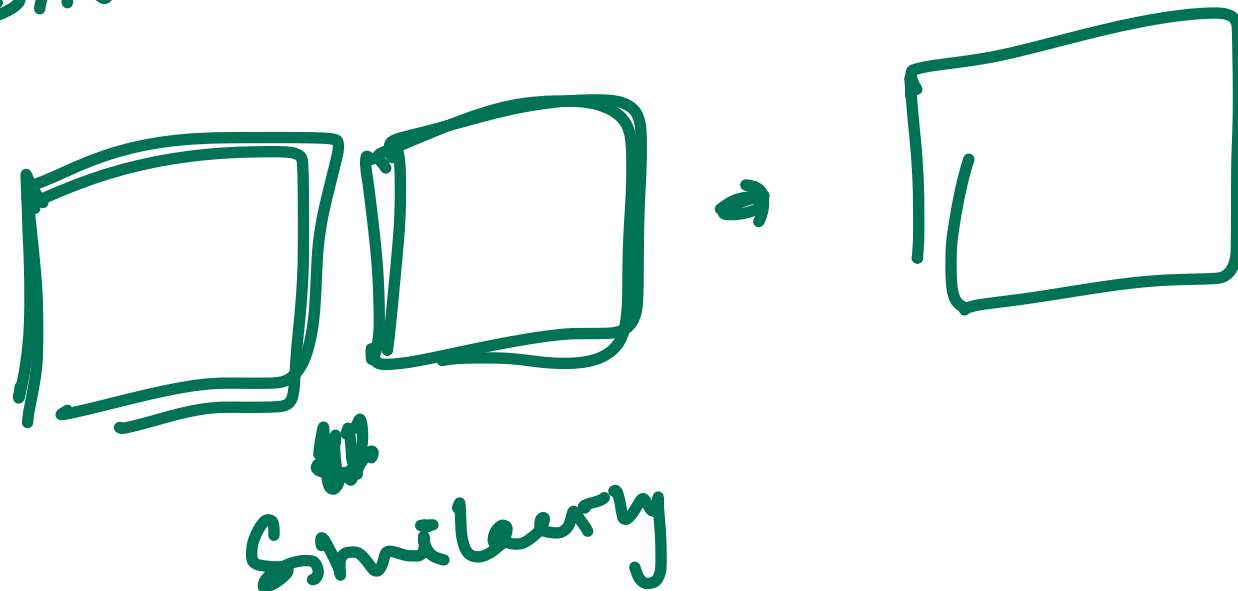
She locked the door with a key

(Trained) → (Content)



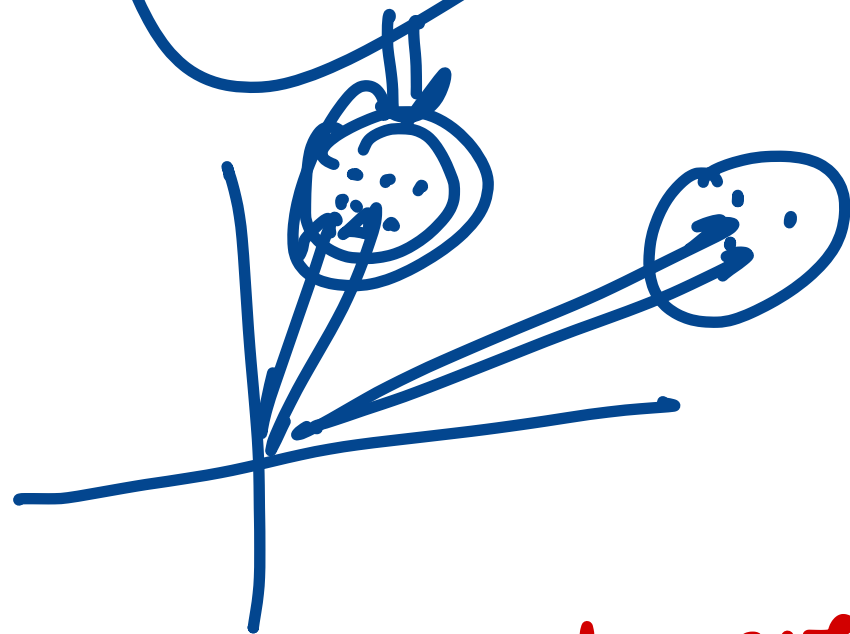


Similar $\rightarrow$ ^{dot} Product of Matrices.



C++
JAVA
Python

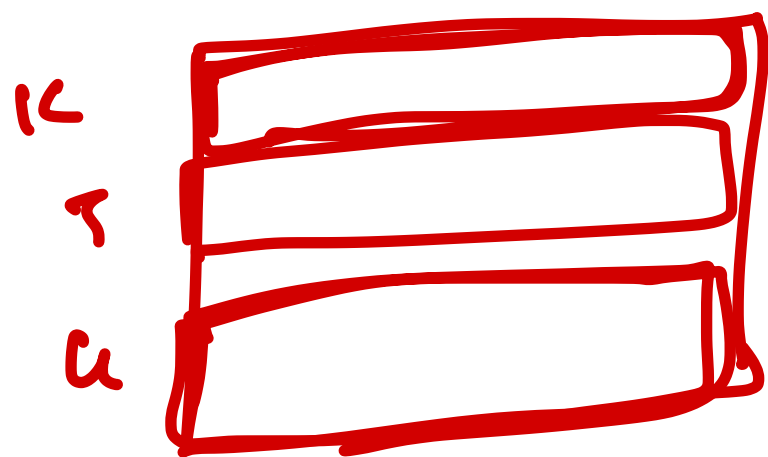
Line by
Review

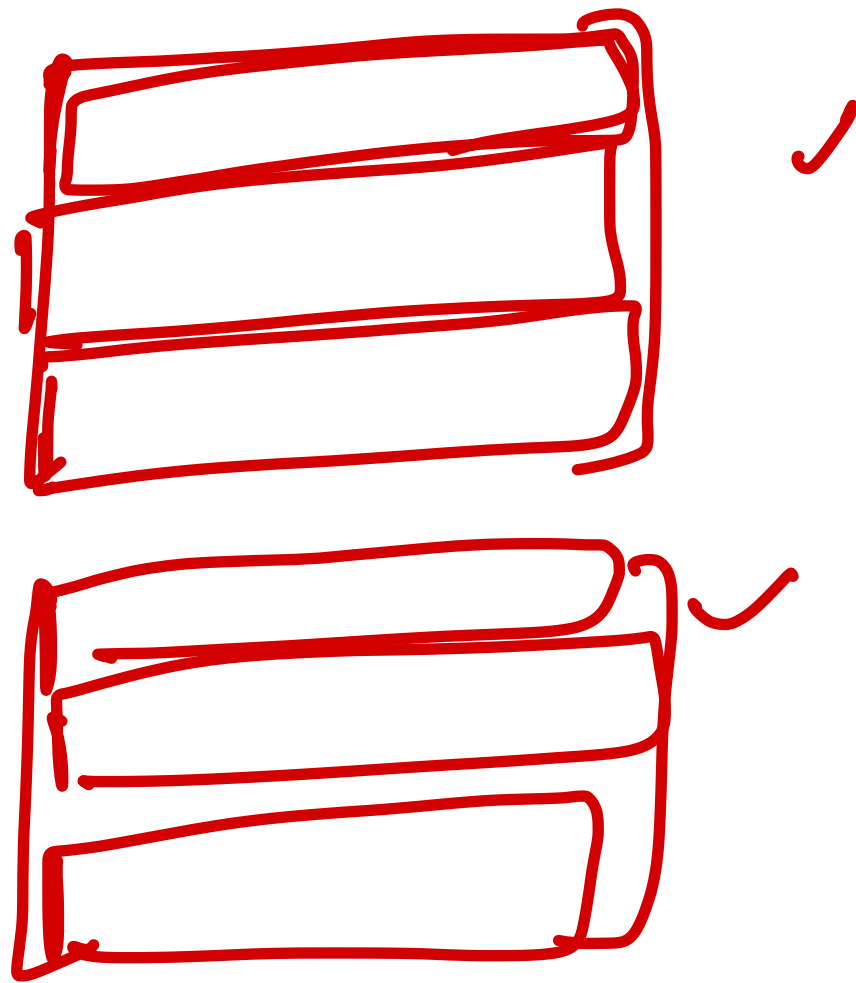


2 vectors for every word / token

1 vector $\rightarrow$ how much does the other tokens have on me

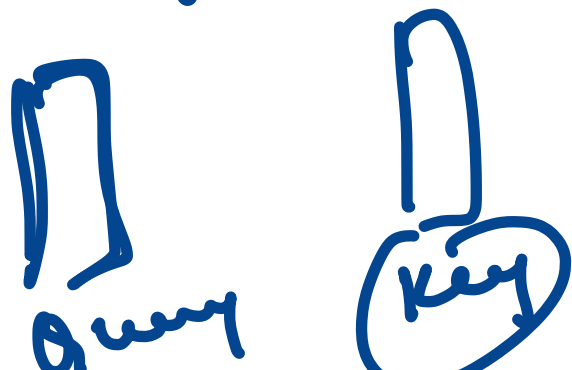
2 vector $\rightarrow$ My impact on you



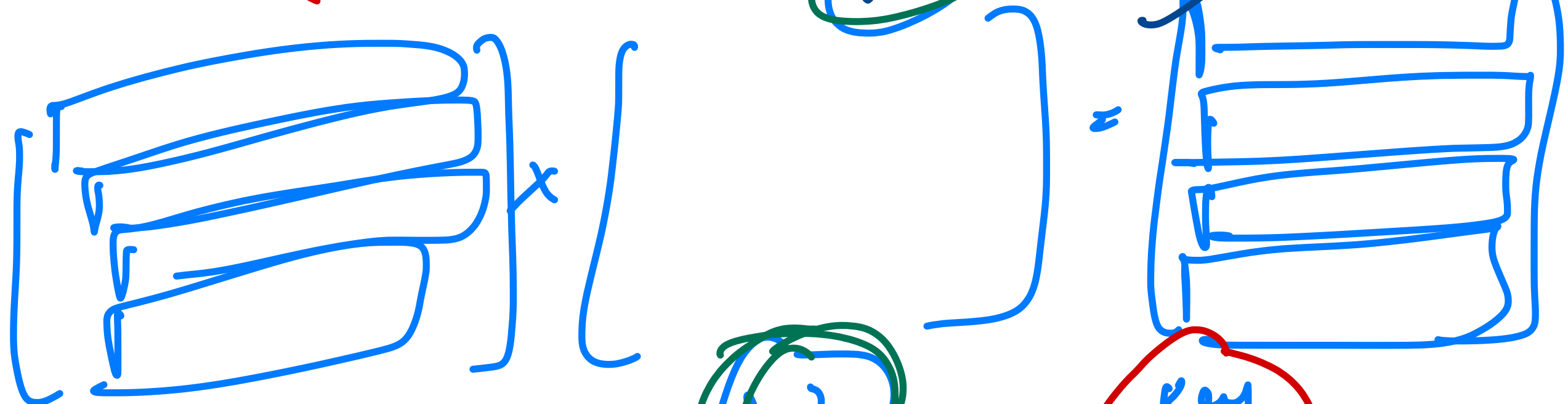
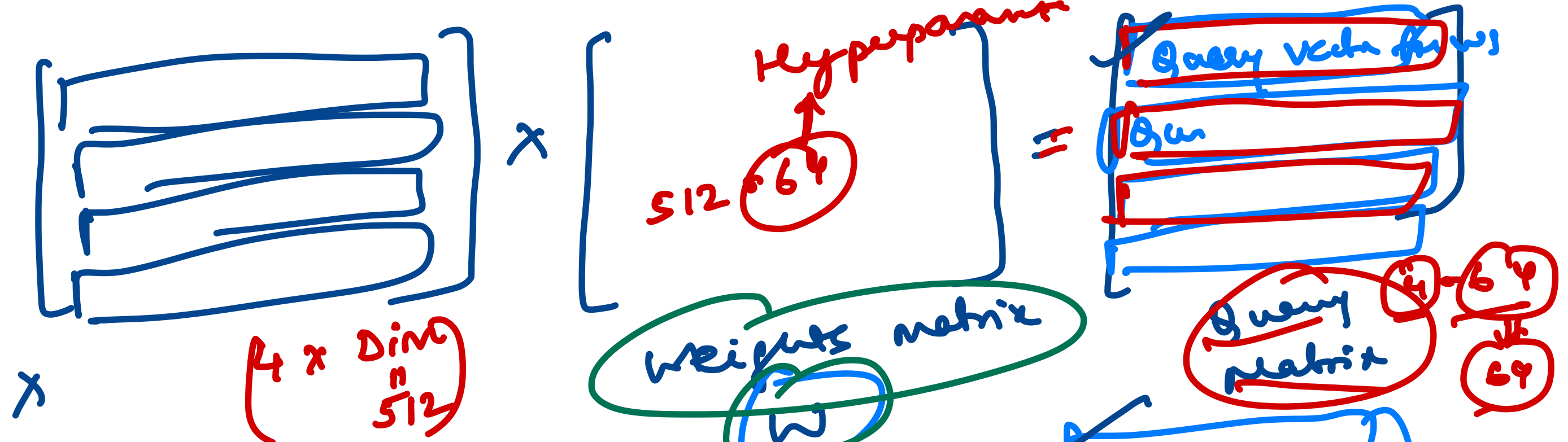


Query key

You Tube
↓
Query [key]



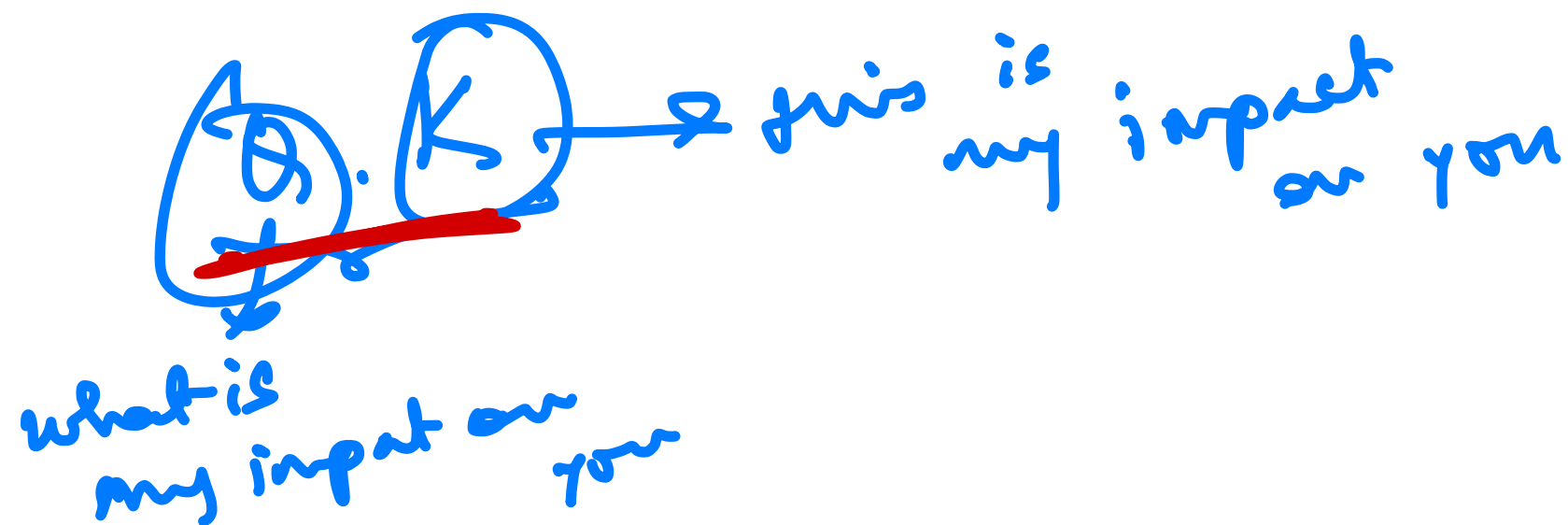
Databases
↓
Query[key]



You is a great bartman
 I love Kohli
 64 dimensions
 64 dimensions

She
unlocked
the
door
with
a
key

<u>Query</u>	<u>key</u>	<u>value</u>	
Q1	K1	V1	Q1, K1
Q2	K2	V2	Q1, K2
Q3	K3	V3	Q1, K3
Q4	K4	V4	Q1, K4
Q5	K5	V5	Q1, K5
Q6	K6	V6	Q1, K6
Q7	K7	V7	Q1, K7
			Q2, K1
			Q2, K2
			Q3, K2



$$(1 \times 64)(64 \times 1)$$

$$Q \cdot K^T \xrightarrow{(512)} (512)$$

$$(4 \times 64)(64 \times 4)$$

no. of words
vocab size

Dimension of
key / query vector

$$0.2 \times 0.3 + 0.5 \times 0.6 + 0.5 \times 0.7 + \dots$$

→ Big

$(Q \cdot K^T) \Rightarrow$ impact / attending / relation /
connection of every token with
every other token

↓

probabilities

sum = 1

$$\alpha = \text{softmax}\left(\frac{Q \cdot k^T}{\sqrt{d}}\right)$$

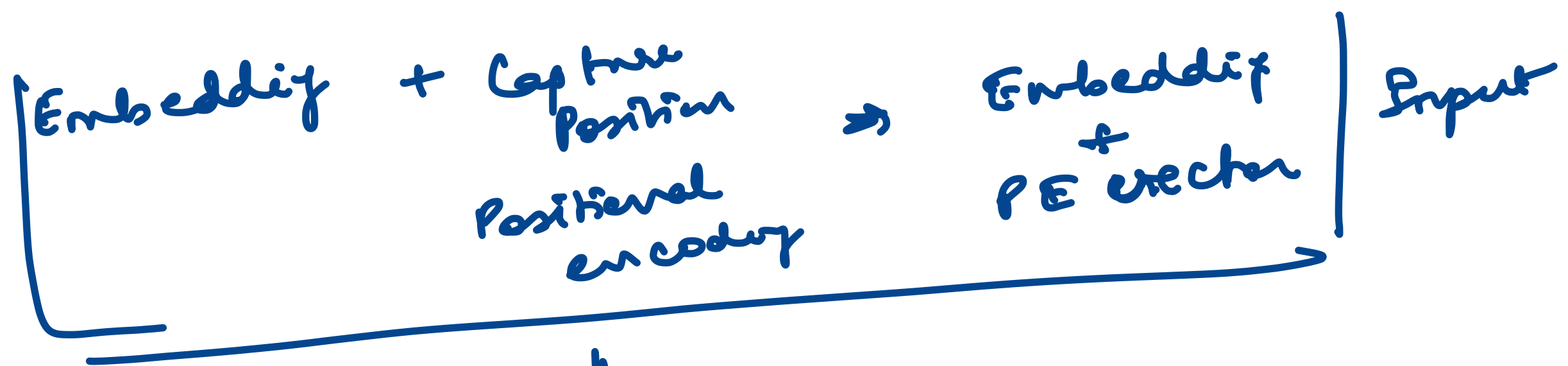
$\Downarrow$

0 and 1

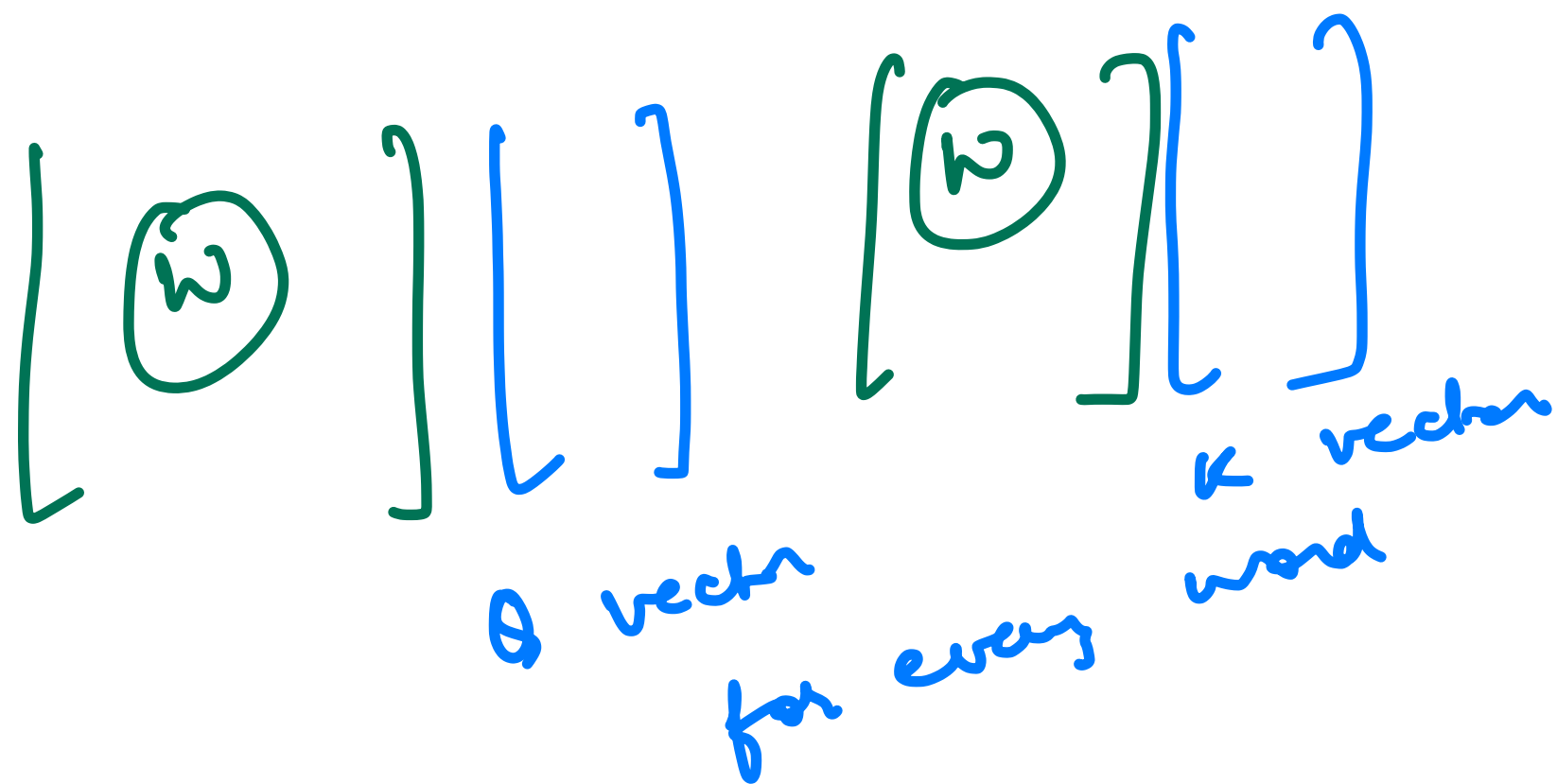
$\rightarrow$ if 2 tokens are really impacting each other

highly relevant
 α will be close to 1.

$\rightarrow$ irrelevant $\Rightarrow \alpha \approx 0$

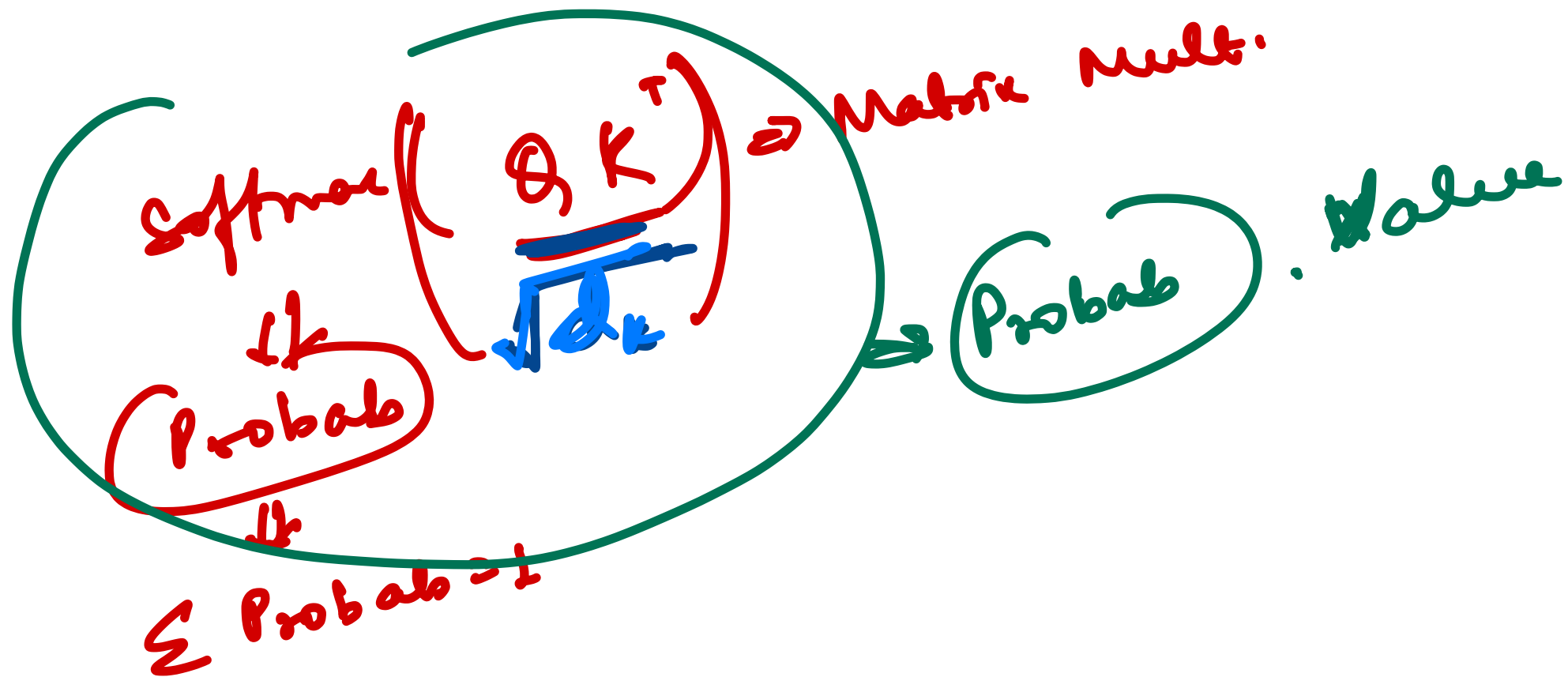


Relation b/w every 2 tokens



Q matrix

Key matrix



$\theta \rightarrow 512$ dimensi
 $K \rightarrow 512$ dimensi

$0.2 + 0.3 + \dots$
512 ~~da~~ time

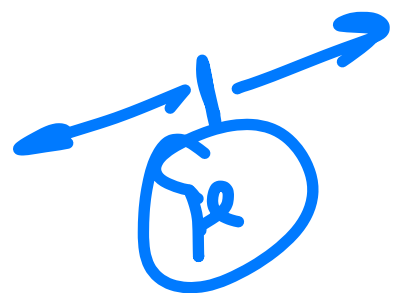
Matrix multiplication

$$a \cdot \textcircled{b} \quad \textcircled{b} \cdot c$$

variance of elements in the matrix

$$2 \cdot \textcircled{3} \cdot \textcircled{3} \cdot 4 = (2 \cdot 4)$$

variance $\propto 3$



$$2 \cdot 8$$

$$8 \cdot 4$$

variance $\propto 8$

$$2 \cdot 4$$

Input $\Rightarrow$

$Q, K^T \Rightarrow$ score

$\Downarrow$
Scaled / Normalized

$\frac{Q, K^T}{\sqrt{d}}$

$\sqrt{d}$

$\Downarrow$

softmax

Probab $\Rightarrow$ 0 and 1

values

Query $\rightarrow$
key $\rightarrow$
value $\rightarrow$

} Dimensions

w_q w_k w_v

Library $\rightarrow$ Transformer
 $\Downarrow$ Query
key $\rightarrow$ Book Name
value $\rightarrow$ Book

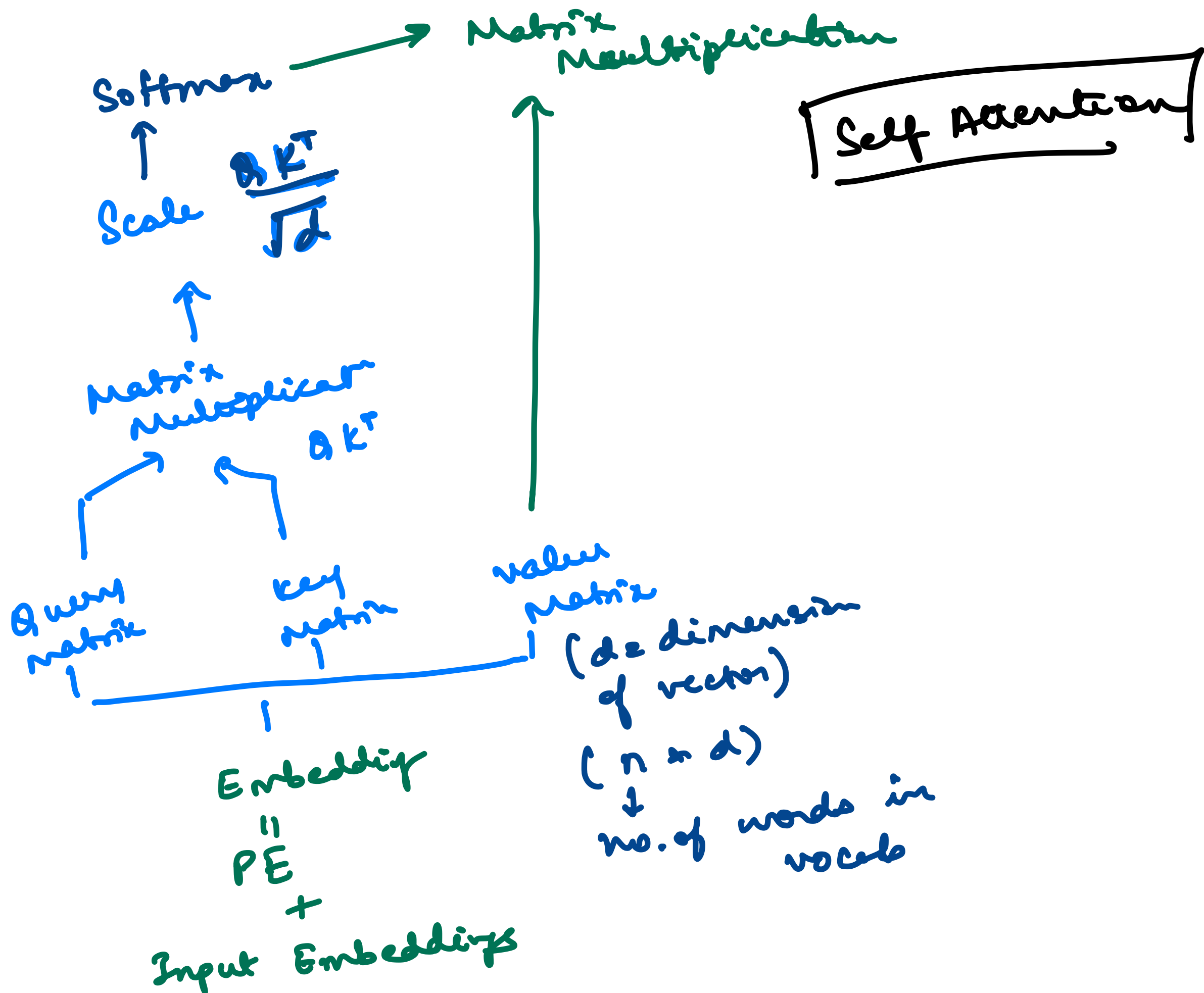
YouTube $\rightarrow$ Query $\rightarrow$ Type
key $\rightarrow$ Title of video
value $\rightarrow$ video.

DB $\rightarrow$ Query.
key $\rightarrow$ ID value $\rightarrow$

Training $\Rightarrow$

W_q, W_k, W_v

To achieve self attention
 $\Rightarrow$ Input fed into 3 distinct models

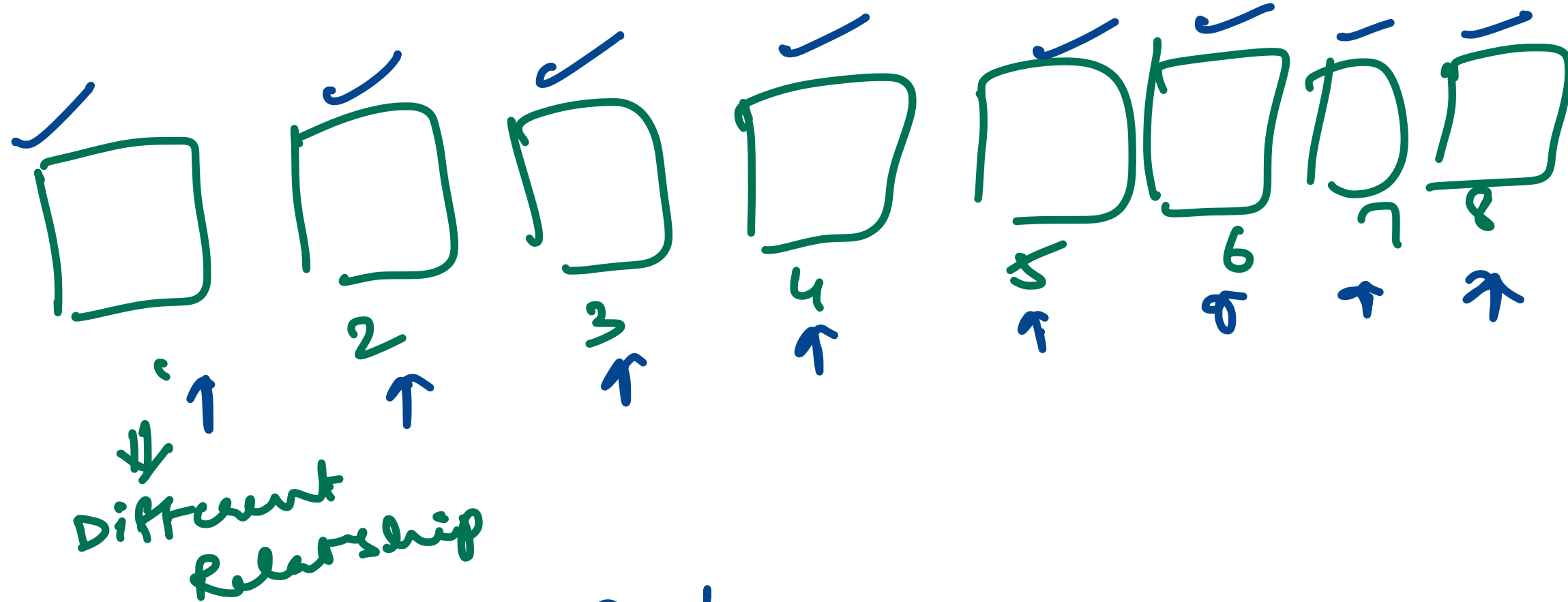


Self Attent $\rightarrow$ Relationship between
every token pair

Naveen gave Satish a beautiful bike
on his bday

(Bike) $\rightarrow$ who is giving?
 $\rightarrow$ when is it being given
 $\rightarrow$ who is being given

eating (apples) $\rightarrow$ verbs
location, time, modifiers



CNN $\rightarrow$ filters $\rightarrow$ (32) / 64
 $\downarrow$
 vertical / hori. / Diagonal.
 $\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \rightarrow$

Encoder $\Rightarrow$ N layers

(6 layers)

Steps Repeated N times

① Input Tokenization $\hookrightarrow$ Embedding
+ Positional Encodings to retain word order

② Multi Headed Attention
 $\rightarrow$ each token is attending to every other token using self attention mechanism

$\rightarrow$ Compute Q, K, V matrices
(learned / training)

$\rightarrow$ Compute Attention Score

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

$$d = \text{Dim}(Q, K, V \text{ vectors})$$

→ Different Relationships
- Multiple Attention heads

$$Q_{matrix} = (\text{embed} + PE) \times (W + \text{matrix})$$

↓
Learning

d dimension

K matrix

V matrix

→ Concatenation layer

Add

③

Normalization layer

④ ⇒ Feed Forward Network
→ 2 Fully connected layers (2 dense layers)
ReLU

⑤ Normalization layers

②.5 $\Rightarrow$ Repeating.